

## Linear Difference Model (1): Autoregressive Exogenous (ARX)

### Model

ARX model:

$$Y(k) = a_1 y(k-1) + a_2 y(k-2) + \dots + a_p y(k-p) + b_0 u(k) + b_1 u(k-1) + b_2 u(k-2) + \dots + b_p u(k-m),$$

*Y is output vector and u is input vector*

The parameters  $a_1, a_2, \dots, a_p, b_0, b_1, \dots, b_m$  are solved using Least Square algorithm.

Above equation can be written in matrix form.

$$\begin{bmatrix} Y(p_1) \\ Y(p_1 + 1) \\ \dots \\ Y(m) \end{bmatrix} = \begin{bmatrix} Y(p_1 - 1) & Y(p_1 - 2) & \dots & Y(1) & u(p_1) & u(p_1 - 1) & \dots & u(1) \\ Y(p_1 + 1 - 1) & Y(p_1 + 1 - 2) & \dots & Y(1) & u(p_1 + 1) & u(p_1 + 1 - 1) & \dots & u(1) \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ Y(m - 1) & Y(m - 2) & \dots & Y(1) & u(p_1 + 1) & u(p_1 + 1 - 1) & \dots & u(1) \end{bmatrix} \begin{bmatrix} a_1 \\ \dots \\ a_{p_1} \\ b_0 \\ \dots \\ b_{p_1} \end{bmatrix}$$

$$\bar{Y} = \bar{A} * par,$$

$$par = pinv(\bar{A}) * \bar{Y}, \text{ where } pinv(\bar{A}) = (A^T A)^{-1} A^T,$$

## Linear Difference Model (2): Autoregressive Moving Average

### Exogenous Model (ARMAX)

Further refinement in predicted values is possible with incorporation of error modeling.

We followed ARMAX algorithm to predict based on error vector of predicted output from ARX and actual output.

By describing error as moving average of white noise we have,

$$Y(k) = a_1 y(k-1) + a_2 y(k-2) + \dots + a_p y(k-p) + b_0 u(k) + b_1 u(k-1) + b_2 u(k-2) + \dots + b_p u(k-m) + c_1 e(k-1) + c_2 e(k-2) + \dots + c_n e(k-n)$$

For m predictions, formulating MATRIX similarly, we can write

$$\bar{Y} = \bar{A} * par,$$

$$par = pinv(\bar{A}) * \bar{Y},$$